

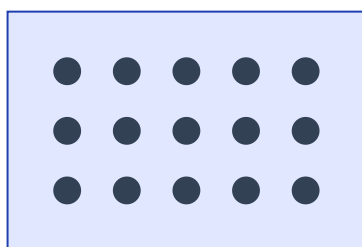
Topic 14 — Particle Model

Edexcel GCSE (9-1) Physics (1PH0) — Paper 2

14.1 States of Matter

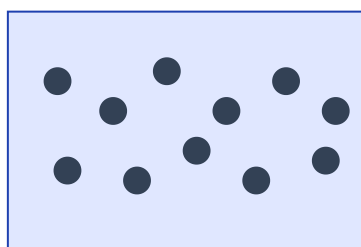
Particle Arrangement in the Three States

Solid



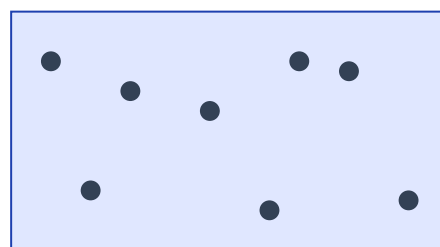
Fixed positions, vibrating

Liquid



Close together, moving past one another

Gas



Far apart, moving rapidly, randomly

14.2 Density

$$\rho = m / V \quad (\text{kg/m}^3)$$

Method for a solid regular shape: measure mass and volume directly. For irregular objects, use a eureka can to measure displaced water volume. For liquids: measure mass of a known volume.

14.3 Internal Energy and Changes of State

Internal energy = total kinetic + potential energy of particles. Heating can increase KE (temperature rises) *or* break bonds during a change of state (temperature stays constant).

$$E = m \times c \times \Delta T \quad (\text{specific heat capacity})$$

$$E = m \times L \quad (\text{specific latent heat})$$

Worked Example: Heat 2 kg of water by 30 °C ($c = 4200$): $E = 2 \times 4200 \times 30 = \mathbf{252\,000\,J}$.
Melt 0.5 kg of ice at 0 °C ($L_f = 334\,000$): $E = 0.5 \times 334\,000 = \mathbf{167\,000\,J}$.

14.4 Gas Pressure and Temperature

Gas particles collide with container walls, exerting pressure. Heating → faster particles → more frequent and harder collisions → higher pressure (at constant V). Compressing at constant T → smaller volume → more collisions per area → higher pressure (**$pV = \text{constant}$**).

$$p_1V_1 = p_2V_2 \quad (\text{for fixed mass, constant } T)$$

Exam Tip: Kelvin temperatures are essential for gas laws: 0 °C $\approx$ 273 K.

Key Points Summary

- Solid → liquid → gas: particles spread out, move faster.
- $\rho = m/V$ — know a method for solids/liquids.
- Specific heat ($E = mc\Delta T$) for temperature change.
- Latent heat ($E = mL$) for change of state — no ΔT .
- Boyle's law: $pV = \text{constant}$ at constant T .